

Wenlin Li

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Ph.D. researcher in optimization, operations research, and machine learning for transportation systems, with a computer science foundation. Interests span large-scale network modeling, transportation planning, decomposition algorithms, and ML for scientific computing. Seeking a Summer 2027 research / applied scientist internship in ML, optimization, or fleet & marketplace decision systems.

Education

Arizona State University

Ph.D. in Transportation Engineering

Tempe, AZ

Expected 2028

University of Utah

B.S. in Computer Science

Salt Lake City, UT

May 2023

Research Experience

Graduate Research Associate — Arizona State University

2024 – Present

- Research in large-scale optimization, operations research, and transportation planning.
- Develop network modeling and decomposition approaches for freight and mobility systems.

Undergraduate Researcher — University of Utah

2022 – 2023

- Research at the intersection of deep learning and scientific computing.
- Improved the efficiency of scientific simulations by combining deep learning with domain knowledge.

Research Projects

Freight Rail Network Optimization — Argonne National Laboratory

2026 – Present

- Contribute to a national-scale freight rail network optimization effort in collaboration with Argonne National Laboratory.
- Develop and validate solver pipelines for large-scale routing and blocking decision support.

Neural PDE Solvers — University of Utah

2024

- Designed CNN architectures to learn first- and second-order spatial derivatives for PDE solvers.
- Integrated learned partial-derivative operators into neural ODE frameworks to improve scientific modeling accuracy.

Skills

- **Programming Languages:** Python, C/C++, MATLAB, Java
- **ML & Scientific Computing:** PyTorch, JAX, scikit-learn, pandas, NumPy; PINNs, neural ODEs, CNNs
- **Optimization & OR:** column generation, Lagrangian relaxation, Branch-and-Price, time-space network modeling, multi-commodity network flow